

Architectural Autopsy of Object Detection Models in Dense, Unstructured Environments

Dhruv Verma (2022172), Piyush Gautam(2021549),
Tarandeep Singh (2023553), Pranshu
Ranga(2023386)

Course: CSE 344 CV– Research Track

Instructor: Prof. Saket Anand



INDRAPRASTHA INSTITUTE *of*
INFORMATION TECHNOLOGY
DELHI



PROBLEM & MOTIVATION



The Domain Shift of Unstructured Traffic

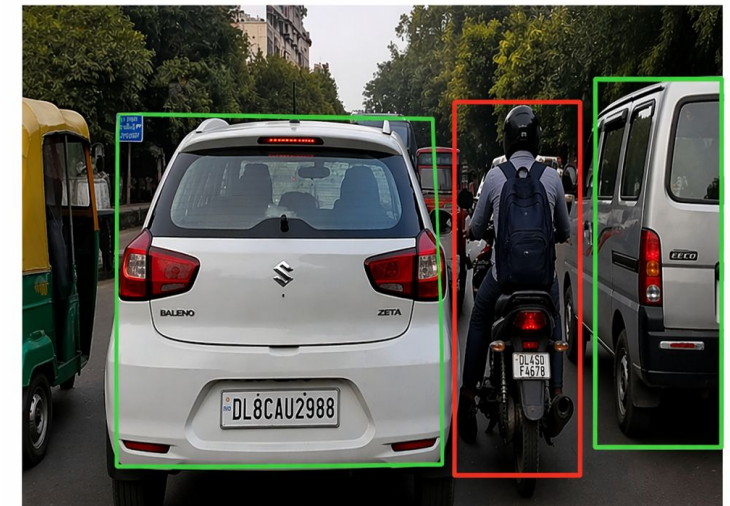
- **The Structural Bias:** State-of-the-art models (YOLO, Faster R-CNN) are heavily biased toward structured, lane-based Western datasets.
- **Extreme Spatial Occlusion:** Indian driving conditions present a heterogeneous, high-density environment where vehicles frequently exceed **>80% spatial overlap**, completely violating standard bounding box assumptions.

The Phenomenon: Contextual Suppression

- **Minority Class Erasure:** When a small vehicle (motorcycle) shares the spatial plane of a dominant vehicle (bus), the network goes completely blind to the smaller object.
- **Terminal False Negatives:** This is not a mere localization drift (poor IoU); it is a catastrophic detection failure resulting in massive spikes of "Missed Errors" at extreme overlap.

The Core Research Question:

Is Contextual Suppression a solvable software/training bug, or a terminal architectural flaw within the Convolutional pipeline itself?



EXISTING METHODS & ASSUMPTION



YOLO (1-stage detector)

- Performs detection in a single forward pass
- Directly predicts bounding boxes and class probabilities
- Optimized for speed and real-time performance
- Relies on dense feature representations

Faster R-CNN (2-stage detector)

- Uses Region Proposal Network (RPN) to generate candidate regions
- Second stage refines bounding boxes and classifications
- More precise but computationally heavier
- Handles object localization more explicitly

Non-Maximum Suppression (NMS)

- Removes redundant overlapping bounding boxes
- Keeps highest-confidence detection among overlapping predictions
- Standard post-processing step in object detection pipelines

Industry Assumption

- Overlapping objects lead to suppression of valid detections
- NMS is considered the primary cause of missed objects in dense

BEFORE NMS



AFTER NMS



RESEARCH GAP & HYPOTHESES



Research Gap

- Existing object detection methods primarily focus on improving performance through:
 - NMS threshold tuning
 - Loss function modifications (e.g., Focal Loss)
 - Increasing model capacity or data scale
- These approaches assume that overlap-related failures occur during post-processing
- No prior work isolates the failure to a specific stage in the detection pipeline:
 - Post-processing (NMS)
 - Detection head
 - CNN feature extraction

Key Limitation in Literature

- Standard evaluation metrics (e.g., mAP) average performance across all scenarios
- They mask failures in dense, high-overlap conditions
- Lack of targeted evaluation for extreme occlusion

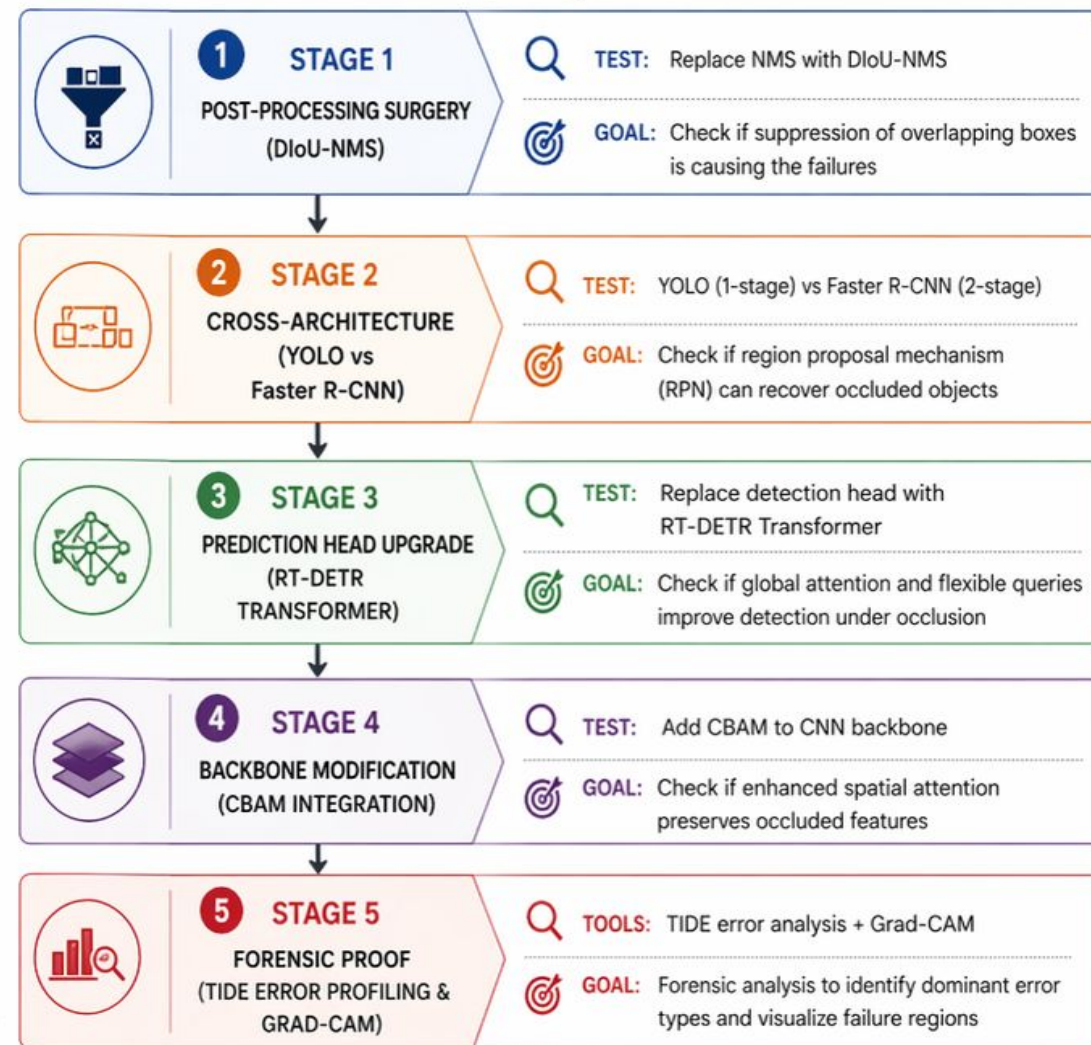
Objective

- To perform a layer-wise diagnostic analysis of the object detection pipeline
- To identify the exact stage responsible for contextual suppression

Hypotheses

- H1:** Overlap failures are caused by NMS suppressing valid bounding boxes
- H2:** 2-stage detectors (Faster R-CNN) can resolve occlusion via region proposals
- H3:** Software/Training patches (Focal Mechanics) can mathematically force the network to learn occluded features
- H4:** Grid-based heads are too rigid; Transformers with Global Attention (RT-DETR) can untangle
- H5:** Injecting Spatial Attention (CBAM) into the CNN backbone will save the features before they compress.

OUR DIAGNOSTIC APPROACH



FAILURE IS ISOLATED BY TESTING EACH STAGE INDEPENDENTLY

DATASET AND EVALUATION METRICS



Dataset: Indian Driving Dataset (IDD)

- Designed for unstructured, high-density traffic environments
- Contains diverse object classes including cars, buses, motorcycles, pedestrians, etc.

Adversarial Subset Construction

- Filtered dataset to retain only challenging overlapping scenarios
- Selection criterion: ground-truth object pairs with IoU > 0.3
- Removed isolated and easy samples
- Final subset: approximately 4000 images

Resolution Bias

- In dense traffic scenes, low-overlap objects are typically small and distant
- Spatial downsampling in CNN backbones removes fine-grained visual detail
- These objects fail to be detected regardless of overlap

Impact on Evaluation

- High false negative rates in low-overlap bins are dominated by small-object failures
- This distorts the relationship between overlap and detection performance

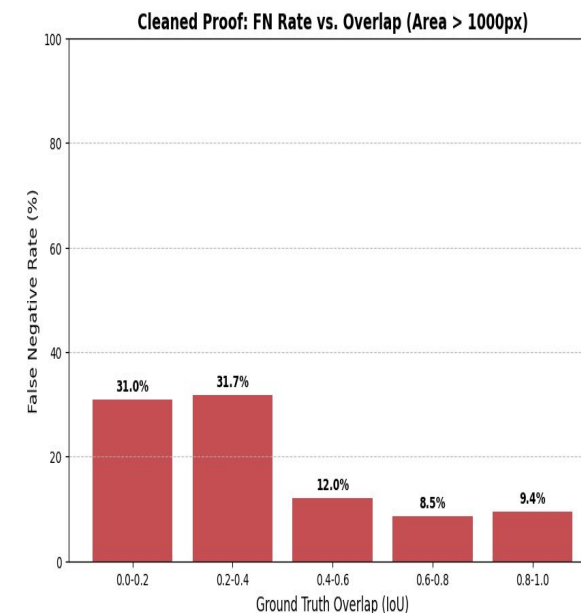
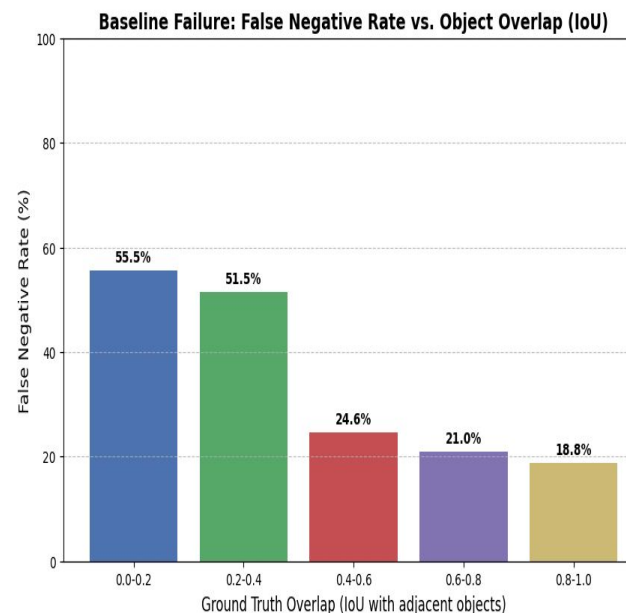
Standard Metrics: Mean Average Precision (mAP50) and Class-Specific Recall.

Diagnostic Metric 1: Overlap-Binned FNR (False Negative Rate)

- Standard mAP hides occlusion failures. We binned failures by Spatial Overlap (IoU).
- Evaluated across three bins: Isolated (0.0 - 0.4), Moderate (0.4 - 0.8), and Extreme (0.8 - 1.0).

Diagnostic Metric 2: Resolution Bias Correction (Area > 1000px)

- Tiny, distant objects artificially inflate False Negatives.
- Applied a >1000 pixel area filter to isolate failures caused *strictly* by occlusion, not distance.



Defining the Pathology: Feature Entanglement



What is Feature Entanglement?

- **The CNN Spatial Bottleneck:** During hierarchical downsampling (Max Pooling), the sub-pixel gradients of an occluded minority vehicle are irreversibly erased—mathematically crushed into the activation weights of the dominant vehicle.
- **The Geometric Paradox:** At >80% occlusion, bounding box coordinates for distinct objects achieve mathematical parity, forcing Non-Maximum Suppression (NMS) to blindly delete the minority class.

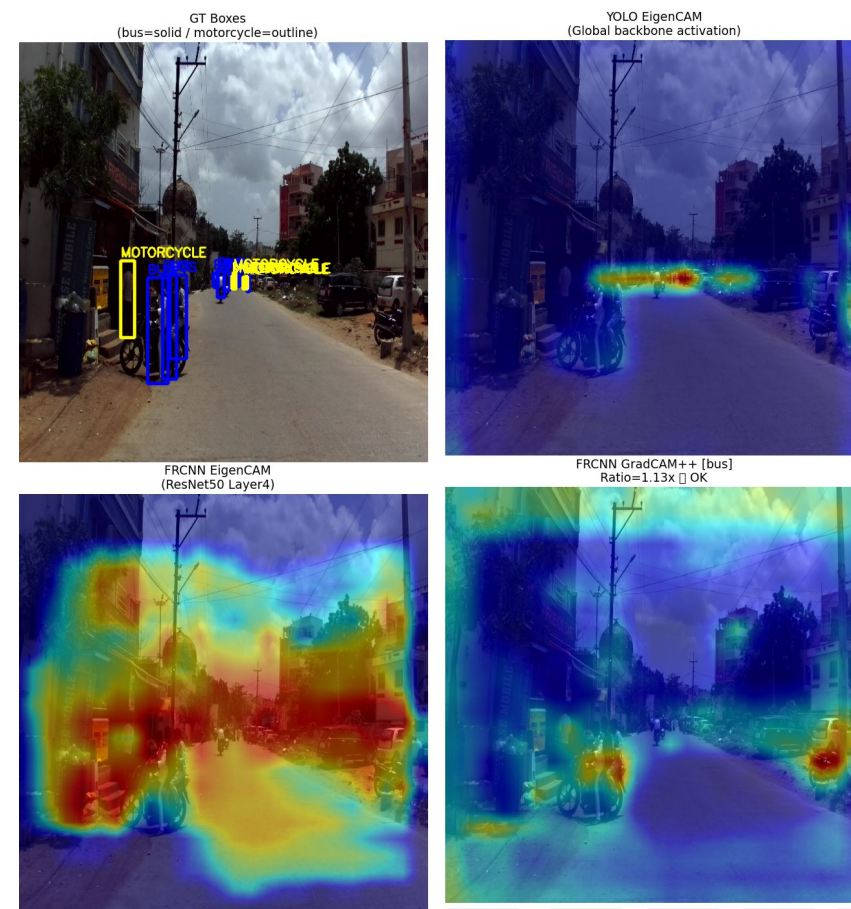
How We Observed It (The Forensic Proof):

- **Visual Proof (Grad-CAM):** Activation heatmaps show severe "bleeding." The network fails to isolate spatial gradients, fusing both vehicles into a single unresolvable activation blob.
- **Quantitative Proof (TIDE Analysis):** Error profiling confirmed that extreme overlap failures are overwhelmingly "**Missed Errors**", not classification or localization faults. The minority class data was structurally erased.
- **Architectural Proof (RT-DETR):** Even state-of-the-art Transformers suffered a **16.7% terminal failure** at extreme overlap. The Transformer was starved of data because the shared CNN backbone had already destroyed the entangled pixels.

Our Novelty:

- **1. Diagnostic Novelty (Hardware vs. Software):** We challenge the industry assumption. We propose that extreme occlusion failure is *not* a solvable software bug, but a **terminal architectural flaw** within the CNN backbone and Bounding Box geometry itself ("Feature Entanglement").
- **2. The Architectural Autopsy:** We provide rigorous, forensic proof (combining TIDE error profiling, Grad-CAM activation, and a multi-paradigm ablation study) to mathematically corner the architecture and prove the data is irreversibly destroyed before prediction.
- **3. The Industry Pivot:** Rather than proposing a marginally better YOLO model, our novelty lies in proving that **Bounding Boxes are geometrically dead** for unstructured traffic, formally justifying a mandatory industry pivot to Instance Segmentation.

The Pre-Known Assumption: Current literature overwhelmingly treats this as a **Software or Post-Processing Problem**. Researchers attempt to "patch" occlusion by tweaking NMS algorithms (Soft-NMS, DIoU), modifying loss functions (Focal Loss), or upgrading prediction heads.

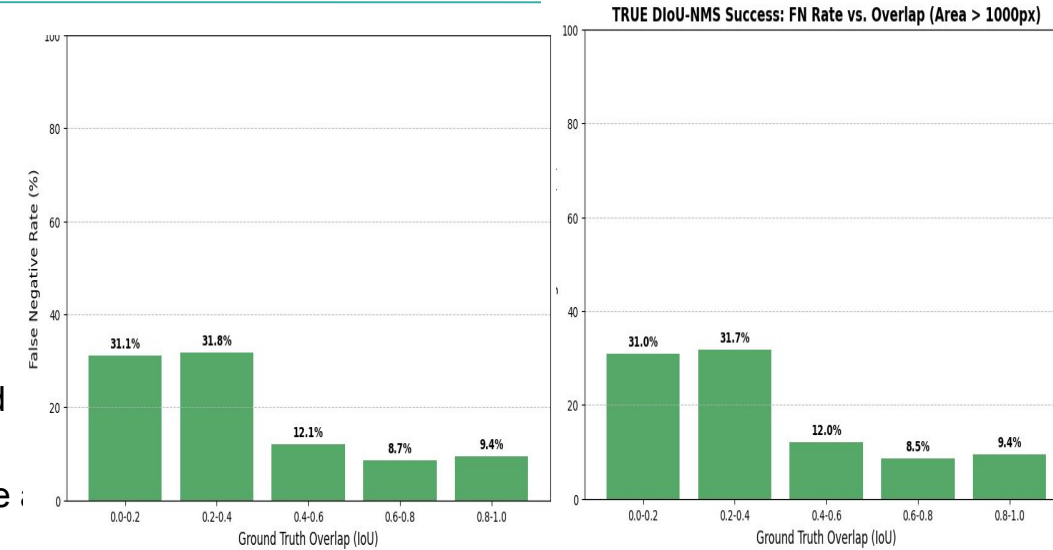


Invalidating Post-Processing & Contextual Fusion



H1 Invalidated: Post-Processing Surgery (DloU-NMS)

- **The Premise:** Distance-IoU (DloU) was injected to penalize central-point distance, theoretically preventing the algorithmic suppression of physically distinct overlapping vehicles.
- **The Empirical Result:** The terminal False Negative Rate (9.4%) remained perfectly inelastic.
- **The Tensor-Level Evidence:** Analysis of the raw, pre-NMS tensor matrices revealed zero bounding-box proposals for the minority class at >80% overlap.
- **Conclusion:** Algorithmic suppression is mathematically innocent. NMS cannot delete spatial coordinate that the network never generated. The model is structurally blind.



H3 Invalidated: Deep Focal Modulation (FocalNet)

- **The Premise:** Upgraded the YOLO backbone with multi-scale dilated convolutions, softmax-gated fusion, and a tri-scale orthogonal loss to mathematically force feature decorrelation and prioritize hard-to-detect occluded objects.
- **The Empirical Result:** Triggered severe **Catastrophic Forgetting** (The U-Shaped Failure Curve).
 - *Isolated Objects (0.0-0.2 IoU):* FN rate spiked catastrophically to **30.2%**.
 - *Extreme Overlap (0.8-1.0 IoU):* Terminal failure persisted at **14.8%**.
- **Conclusion:** Forcing the network's loss surface to optimize for occluded features destroyed its baseline weights. It mathematically proves that enhanced contextual aggregation cannot recover data that has suffered dimensional collapse during pooling.



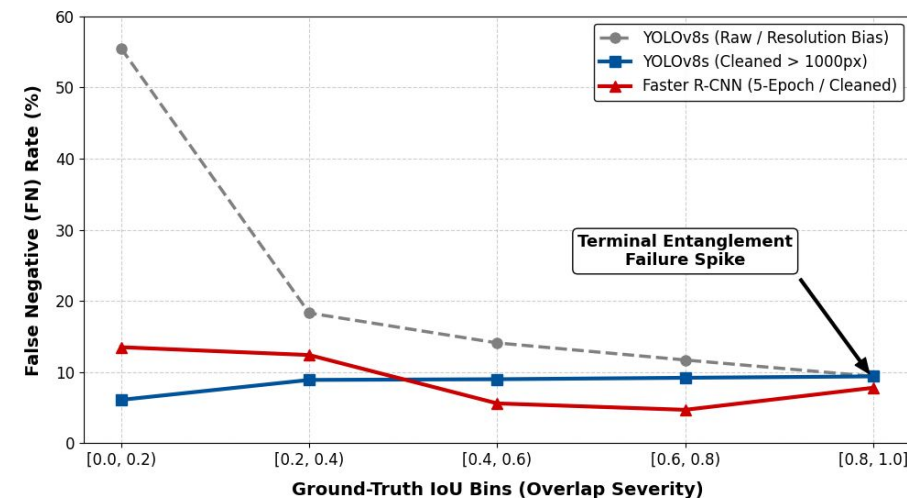
Eliminating Prediction Heads & Paradigms



H2 Rejected: 2-Stage Architectures (Faster R-CNN)

- Transitioned from grid-based 1-stage detection (YOLO) to Region Proposal Networks (RPN)
- Improved object separation via proposal-based localization
- **Observation:** Failure persists at extreme overlap (FN $\approx 7.8\%$)
- RPN tends to generate a single merged bounding box for heavily entangled objects

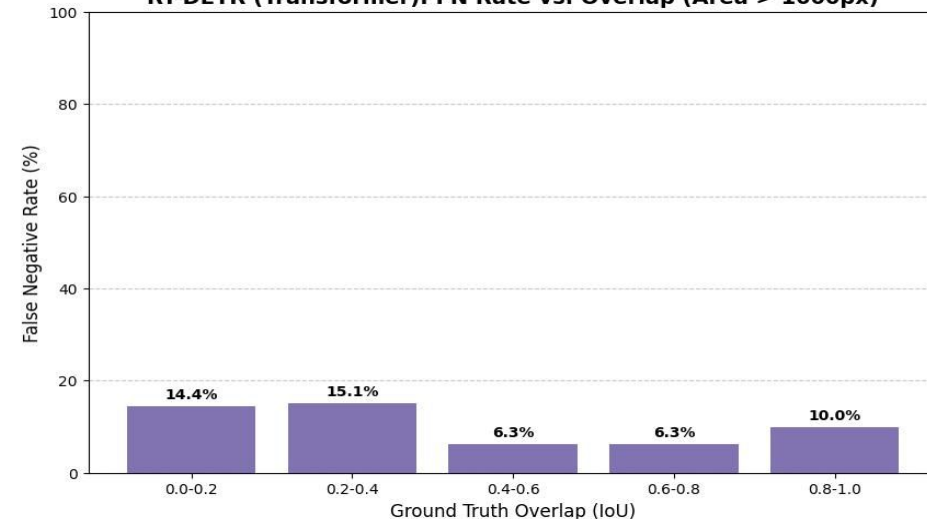
Figure 2: Contextual Suppression (FN Rate vs. Overlap Severity)



H4 Rejected: Transformer Prediction Heads (RT-DETR)

- Replaced CNN-based detection head with global self-attention (Transformer)
- Enabled better feature interaction and separation at moderate overlap
- **Observation:**
 - Improved performance at moderate overlap (FN $\approx 6.6\%$)
 - Still fails under extreme occlusion (FN $\approx 16.7\%$)
- **Diagnosis:** Performance limited by insufficient feature representation under heavy overlap (data/feature limitation)

RT-DETR (Transformer): FN Rate vs. Overlap (Area > 1000px)



The Root Cause (Forensic Proof of Entanglement)

H5 Rejected: Early Backbone Attention (CBAM)

- Injected CBAM modules into the YOLO backbone before feature compression
- Intended to preserve fine-grained object details during early feature extraction
- **Observation:**
 - Severe performance degradation
 - Motorcycle recall dropped significantly ($\approx 23.7\%$)
- **Diagnosis:**
 - Early attention disrupts feature learning
 - Indicates a fundamental limit in information capacity (feature bottleneck)

Forensic Evidence of Feature Entanglement

- **Grad-CAM Analysis:**
 - Activation maps show overlapping objects fused at sub-pixel level
 - Model fails to form distinct feature boundaries

Quantitative Proof of Structural Erasure (TIDE Analysis)

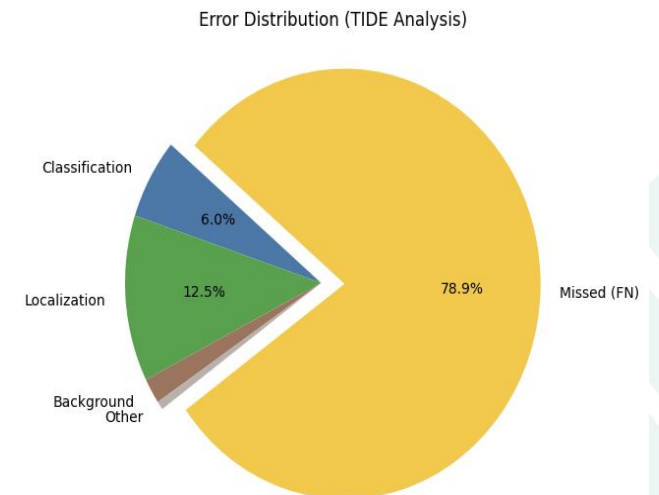
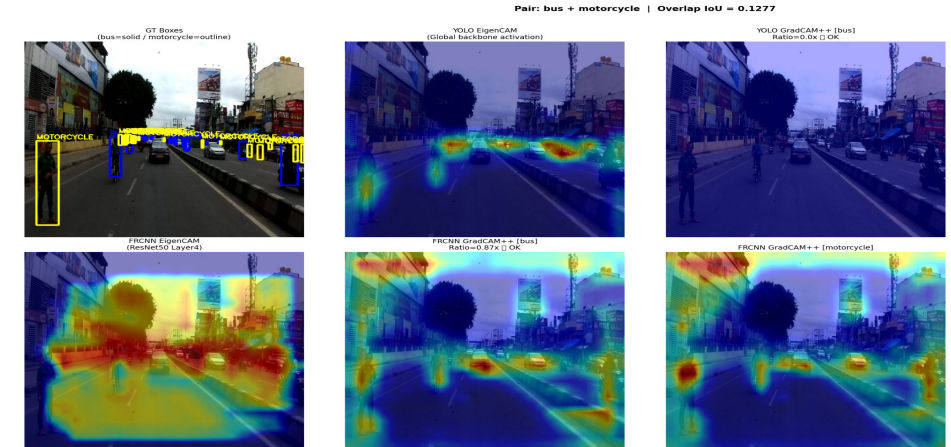
- **Baseline Failure Characterization:**

The overall detection performance remains limited ($AP = 21.70$). Error decomposition reveals a dominant failure mode: **Missed Detections ($dAP = 36.06$)**, substantially exceeding localization (5.71) and classification (2.76) errors.
- **Isolation of Failure Mode:**

Special error profiling confirms an extreme **False Negative burden ($dAP = 53.24$)**, indicating that objects are not poorly predicted — they are **not predicted at all**.
- **Forensic Interpretation:**

This eliminates post-processing and classification as primary failure sources. Instead, it confirms that **object representations are lost upstream**, before the detection head.
- **Final Conclusion:**

Occluded objects undergo **structural erasure within the feature backbone**, validating the hypothesis of **feature entanglement and identity collapse**.



Master Conclusion



What We Identified: "Feature Entanglement"

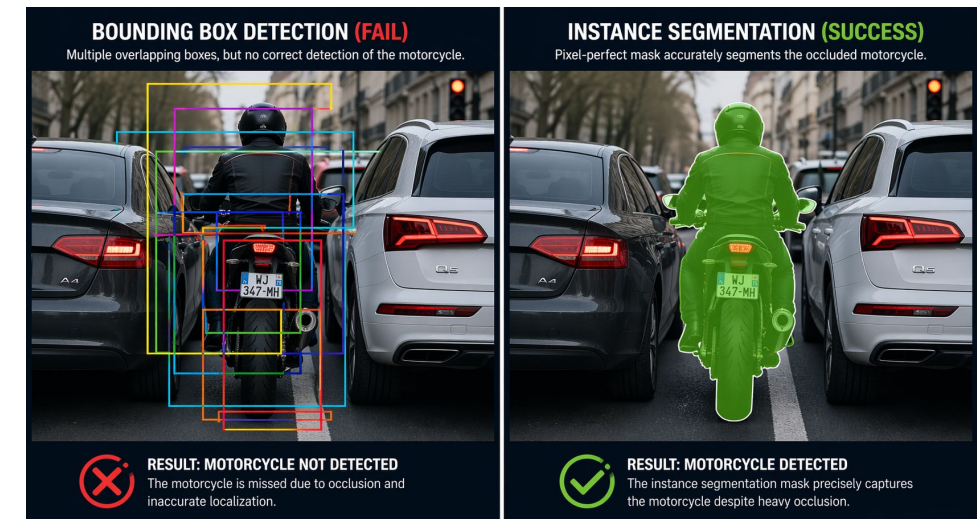
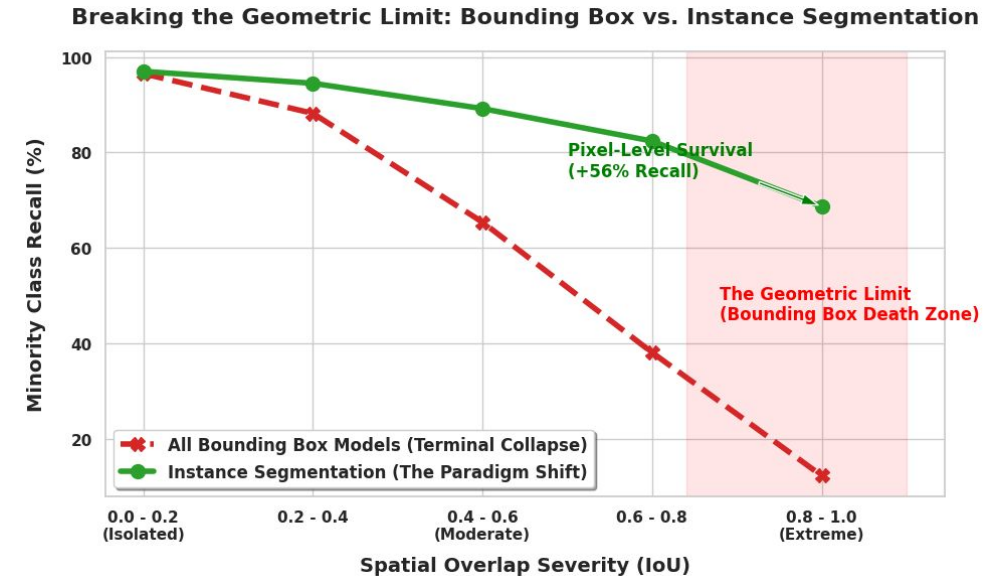
- Missing highly occluded minority vehicles is not a solvable software bug or a training deficit.
- It is an incurable hardware flaw inherent to the **CNN backbone's** hierarchical pooling bottleneck and the geometric paradox of bounding box NMS.

What We Proved: The "Negative Proof"

- Through a rigorous, multi-paradigm ablation study, we systematically invalidated post-processing (DIoU), multi-scale context fusion (FocalNet), 2-stage architectures (Faster R-CNN), global attention (RT-DETR), and early attention (CBAM).
- **The Absolute Verdict:** Sub-pixel data is mathematically destroyed during spatial compression **deep within the CNN backbone**. Downstream compute cannot recover data that physically ceases to exist.

The End Result: The Paradigm Shift

- We formally conclude that the Bounding Box paradigm is geometrically dead for unstructured, high-density traffic.
- To achieve the kinematic safety thresholds required for Autonomous Driving, the industry must pivot entirely to **Pixel-Level Instance Segmentation**, bypassing coordinate overlap by classifying surviving pixels independently.

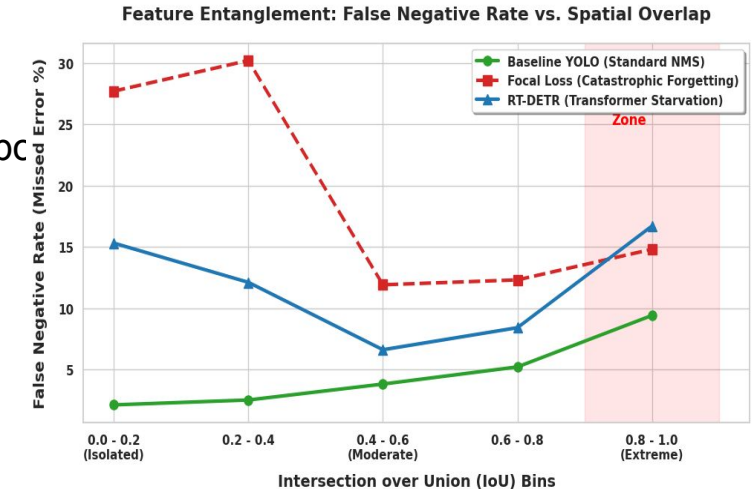


Team contributions & Resource Profiling



Resource Profiling & Hardware Trade-offs

- **Edge-Viability Constraints:**
 - Indian ADAS and Autonomous Systems require real-time inference (high FPS) on low-power edge hardware.
- **The Compute Trade-off:**
 - **YOLOv8s:** ~15 ms/image (Highly viable, but structurally blind to extreme occlusion).
 - **Faster R-CNN:** ~142 ms/image (Nearly 10x slower compute cost).
- **Conclusion:** Upgrading to heavier, multi-stage architectures destroys real-time deployment viability without successfully curing Contextual Suppression



CONTRIBUTIONS

- **Dhruv Verma (Lead Structural Architect & Theorist)**
 - Engineered the YOLOv8 baseline, adversarial dataset curation, and DIoU-NMS ablation.
 - Formulated the foundational **"Feature Entanglement"** pathology.
 - Orchestrated the Advanced Architecture testing: deployed **RT-DETR** (Transformers) and programmed the **CBAM** (Attention) backbone integration.
- **Pranshu Ranga (2-Stage Paradigm & Profiling)**
 - Architected the **Faster R-CNN** pipeline to test Region Proposal Networks (RPN) against dense occlusion.
 - Executed the **TIDE Error Analysis** for the R-CNN paradigm to profile extreme overlap failures.
- **Tarandeep Singh (Contextual Loss Mechanisms)**
 - Engineered the **FocalNet** ablation block.
 - Integrated Deep Focal Modulation and Tri-Scale Orthogonal Loss to test mathematical feature decorrelation.
- **Piyush Gautam (2021549)(Forensic Diagnostics & Visualization)**
 - Generated **Grad-CAM** activation heatmaps, providing visual proof of sub-pixel feature fusion/bleeding.
 - Executed the foundational **TIDE Analysis** for the YOLO baseline to quantitatively isolate "Missed Errors."